



نظم التشغيل (أهليه)

9:30-10:30

24/11/2024

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Faculty of Computers and Artificial Intelligence, Assiut National University

Course code: CS321

3rd Level

Mid-term Exam

Duration: one hour

This exam for the following program(s) : (Add the program's name(s))

Points: -/20

1

الإسم الرباعي (بالعربي فقط) *



2

رقم الجلوس *



المستوي * 

- ☐ الأهلية علوم الحاسب
- ☐ الأهلية نظم المعلومات
- ☐ الأهلية هندسة البرمجيات

○ الأهليلة الذكاء الإصطناعى

☐ ٣هـ

☐ ٤أ

☐ ٤ب

☐ ٤ج

☐ ٤د

☐ ٤هـ

6

رقم الكمبيوتر * 

2

7

الكود * (قد تمت مراجعة بيانات الطالب ورقم الجلوس) 

COLJ

 **Will be reviewed**

8

What is a short-term scheduler? 

☐ It selects which process must be brought into the ready queue

- ☒ It selects which process must be executed next and allocates CPU
- ☐ It selects which process must be removed from memory by swapping
- ☐ None of the mentioned

 **Will be reviewed**

9

The primary function of the Bootstrap program is to: 

- ☒ Initialize the system and load the operating system
- ☐ Manage user applications
- ☐ Connect to the internet
- ☐ Run games and multimedia

 **Will be reviewed**

10

Which type of process would benefit most from a high-performance CPU? 

- ☐ I/O-bound process
- ☒ CPU-bound process
- ☐ Both equally
- ☐ Neither

 **Will be reviewed**

11

What is the role of the interrupt vector table in handling interrupts? 

- ☐ Controls the speed of the CPU
- ☐ Stores the location of all programs in memory
- ☒ Maps interrupt types to their corresponding interrupt service routine (ISR) locations
- ☐ Defines memory access permissions

 **Will be reviewed**

12

Suppose that a process is in "Blocked" state waiting for some I/O service. When the service is completed, it goes to the _____ 

- ☐ Terminated state
- ☒ Ready state
- ☐ Running state
- ☐ Suspended state

 **Will be reviewed**

13

The Bootstrap program resides in: 

- ☐ RAM
- ☒ Read-Only Memory (ROM)
- ☐ User Space
- ☐ Hard Disk

 **Will be reviewed**

14

What is the main purpose of Direct Memory Access (DMA)? 

- ☐ To manage memory allocation for user programs
- ☒ To transfer data between I/O devices and memory without CPU intervention
- ☐ To provide a user interface for interacting with the system
- ☐ To schedule processes for execution

 **Will be reviewed**

15

For the following system of four processes with two resources (R1, R2): If the availability vector is [2 4], is the system deadlocked? 

- ☐ Yes
- ☒ No

 Will be reviewed

16

What is a long-term scheduler? 

- ☒ It selects processes which must be brought into the ready queue
- ☐ It selects processes which must be executed next and allocates CPU
- ☐ It selects processes which must be removed from memory by swapping
- ☐ None of the mentioned

 Will be reviewed

17

Which of the following scheduling algorithms gives the minimum average waiting time? 

- ☐ FCFS
- ☐ Priority
- ☐ Round – robin
- ☒ SJF

 Will be reviewed

18

Round robin scheduling falls under the category of _____ 

- ☐ Non-preemptive scheduling
- ☒ Preemptive scheduling
- ☐ All of the mentioned
- ☐ None of the mentioned

 **Will be reviewed**

19

A process is in the "Ready" state when: 

- ☐ It has finished executing
- ☐ It is currently executing
- ☐ It is being created
- ☒ It is ready to execute but the CPU is currently assigned to another process

 **Will be reviewed**

20

The longer the time slices, the more a RR scheduler gives similar results to a FIFO scheduler 

- ☒ True
- ☐ False

 **Will be reviewed**

21

Assuming the processes is being scheduled with the SJF scheduling algorithm (non-preemptive). The waiting time for process C is: 

- ☒ 4
- ☐ 6
- ☐ 8
- ☐ None of the above

 **Will be reviewed**

22

The real difficulty with SJF in short term scheduling is _____ 

- ☐ it is too good an algorithm
- ☒ knowing the length of the next CPU request
- ☐ it is too complex to understand
- ☐ none of the mentioned

 **Will be reviewed**

23

DMA is used for _____ 

- ☒ High speed devices
- ☐ Low speed devices
- ☐ Utilizing CPU cycles
- ☐ All of the mentioned

 **Will be reviewed**

24

The interval from the time of submission of a process to the time of completion is termed as _____ 

- ☒ turnaround time
- ☐ response time
- ☐ waiting time
- ☐ throughput

 **Will be reviewed**

25

Assume Quantum value two, given a RR scheduler, what is the response time of process B? 

- ☐ 0
- ☒ 1
- ☐ 2

☐ 3

☐ None of the above

 **Will be reviewed**

26

For the following system of four processes with two resources (R1, R2): If the availability vector is [2 3], is the system deadlocked? 

☒ Yes

☐ No

 **Will be reviewed**

27

Given a FIFO scheduler, what is the average response time for the four processes? 

☐ 4.5

☒ 5

☐ 6.5

☐ 8

☐ None of the above



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